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Research Alignment Workshop II

Human-Centered Adoption of Large Language Models Towards Cross-Domain Evidence, Comparative Studies, and Systematic Adoption Frameworks

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From technical success to sustainable adoption

LLM research tells us increasingly well what models can do

We know much less about:

- what happens when LLMs enter real workflows;
- how people adapt to them;
- how responsibilities and roles change;
- which organizational conditions matter;
- which findings generalize across domains.

What makes an LLM application adoptable — not merely technically successful?

From MIPRO to SOFTCOM

- **MIPRO 2026**
Initial framework
Pilot Workshop
11 structured responses
Expert Reflection
HRI + SE/trustworthy AI perspectives
- **SOFTCOM 2026**
Cross-domain cases + framework refinement
Extended Case Studies
Cross-domain synthesis
- **This is not a repeated workshop.**
- **SOFTCOM is the second empirical iteration of the research initiative.**

What are we trying to build?

1. Common reporting structure

How can LLM adoption cases be described comparably?

2. Cross-domain analytical framework

Which challenges, benefits, and conditions recur?

3. Evidence-based adoption guidance

What can organizations learn and reproduce?

Today we are primarily working between Levels 1 and 2.

Human-Centered Socio-Technical Perspective

LLMs are not isolated tools

- They operate within:
 - human workflows
 - organizational structures
 - regulatory constraints
 - collaborative decision-making
 - software-intensive ecosystems

Human-centered adoption requires:

- trust
- transparency
- human agency
- accountability
- sustainable integration

Proposed Analytical Framework

- **9 Core Dimensions for Analysis across domains**

- A Problem - What is being solved?
- B Solution - How is it solved?
- C Human Interaction - How do humans engage?
- D Context - Under what conditions?
- E Evaluation - What is the effect?
- F Challenges - What limits it?
- G Benefits - What value is created?
- H Transferability - What generalizes?
- I Process - How to reproduce it?

Goal: comparable and cumulative knowledge across domains

Not a final framework. A common reference point for comparison and discussion.

A question emerging after MIPRO: **Are we human-centered enough?**

System-centered question

- How do humans interact with the system?

Human-centered questions

- What do people need?
How do they experience the system?
What agency and control do they retain?
How does AI change their roles and practices?
Who benefits — and who bears the cost?

Is “human-centered” one dimension — or a lens across all nine dimensions?

What we ask from today's cases

Do not ask only: Does the LLM work?

We Ask also:

- For whom does it work?
- Under which conditions?
- What changes for humans and organizations?
- What limits adoption?
- What value is created — and for whom?
- What can transfer to another context?
- What would be needed to reproduce the adoption?

We do not expect every case to answer all nine questions. Missing answers are also evidence.

II. CASE STUDIES

14:40–15:15 (35 min)

Cross-Domain Case Studies

- **Case 1 — Industrial Cybersecurity**
Enhancing Operation Wisdom Logging Solution Utilizing Different Local LLMs for Security Anomaly Detection
Zoran Bosić — Ericsson Nikola Tesla
- **Case 2 — Edge LLM Infrastructure**
Payload-Aware Dynamic Precision Orchestration for Edge LLM Inference
Dr. Rana Abu Bakar
- **Case 3 — Network Management & Reliability**
Network Management Reliability Enabled by LLM: Analysis of Channel Capacity Case in the Presence of Beaulieu-Xie Fading
Dragana Kisić — University of Niš
- **As you listen, ask yourself:**
What is common? What is context-specific? What is missing from our framework?

III. STRUCTURED REFLECTION

oko 15:15 (10 min)

Individual Reflection

- **Now map what you heard — and your own case**
- **Please complete the reflection sheet individually.**
 - There are no “correct” answers.
 - Think about your own LLM case if you have one.
 - Otherwise, reflect on one of the presented cases.
 - Mark what the framework captures.
 - Note what it misses.
- The link to the reflection sheet is at workshop web page
- <https://tgalinacgrbac.github.io/workshop/>

One additional reflection

- **What is missing?**
- Does the framework sufficiently capture the **human perspective**?
- Are any questions **too broad to produce comparable answers**?
- Is there an important dimension that is **not represented at all**?

IV. MODERATED CROSS- DOMAIN DISCUSSION

15:25 (20 min)

Across the Three Cases

What patterns recur?

- **COMMON**

What appeared across cases?

- **CONTEXT-SPECIFIC**

What depended on domain or environment?

- **MISSING**

What was important but not captured by the framework?

Human-Centered Lens

What changes when we look from the human perspective?

- Who uses or is affected by the system?
- What control remains with humans?
- Where is human judgment essential?
- How are roles and practices changed?
- Who gains value?
- Who carries additional burden or risk?

Human Interaction as one dimension — or human-centeredness across all dimensions?

From Performance to Adoption

- **Technical capability**
- **usefulness**
- **human acceptance**
- **workflow integration**
- **organizational adoption**
- **sustained use**

Where does each of today's three cases currently sit?

V. FRAMEWORK REFINEMENT

15:45 (10 min)

Keep, Refine, Add?

KEEP

- What works well?

REFINE

- What needs clearer operationalization?

ADD

- What is missing?

VI. FUTURE RESEARCH

15:55–16:00

What Comes Next?

From workshops to cumulative evidence

- **MIPRO**
Pilot dataset
- **SOFTCOM**
Cross-domain expansion & refinement
- **Extended Case Studies**
Structured reporting using the analytical framework
- **JCOMSS Special Issue**
Comparative empirical evidence
- **Cross-Domain Synthesis**
Evidence-based human-centered LLM adoption framework
- **Interested in contributing?**
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